

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1220

Glacier 230 kV
June 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1220. The Interconnection Customer proposes a 162.7 MW wind facility interconnecting at the Glacier 230 kV substation in the Ohio area. The request was received on June 05, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 312 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,183,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Lawrence Energy Center Substation-Barna 100.0 kV: 107.7% loading (143 MVA) following the outage of the Bayport Sw Sta-Firstenergy Bay Shore 230.0 kV; DFAX = 7.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barna-Lawrence Energy Ctr 100.0 kV Line	Reconductor line	\$3,183,000	100% (\$3,183,000)
	Total allocated to project		\$3,183,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.